

Project Proposal

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The aim of the project is to design, implement and train a latent diffusion model that maps a robotic hand grasp to a possible shape that that grasp is grasping. This problem can be important in an industrial setting where the robotic arms are not that complex and expensive but simple and cheap. In this setting a vision input (such as a camera) is not available or if available not accurate to many reasons (occlusion). The goal of the robotic system would be to get the idea of the object that it grasped so that it could better manipulate or move the object through the environment collision-free.

There has been not an abundant literature on this topic, probably because of the focus and dominance of the vision input system. One interesting approach to this topic is trying to model a joint distribution of the grasp, shape and contact points [1] or another approach by [7,8,9]. While an interesting approach, we would like to try only to model the conditional distribution with respect to the grasp data, without calculating the contact points, hoping to make a faster and more lightweight model.

The main idea of our model is to have a pretrained variational autoencoder together with a diffusion model, where the VAE would be used to compress the 3D problem to a smaller latent space. Experiment with various VAEs [2,3,4,5,6]. In this latent space the diffusion and denoising process would take place. The diffusion model would be trained and conditioned on the grasp input. The diffusion model would be mainly using attention blocks. One idea is to impose a structure of the hand and the object in such a way that only the connected parts talk to each other. We will experiment with connecting the object position to the object coordinates and object coordinates to certain parts of the hand, basically trying to enforce a graph-like knowledge to our network. Of course, the latent code that is being denoised would be talking to all the parts of our system (fingers, object position, object pose). Since the input data is very scarce, we could try to calculate a point cloud of our hand using the hand model and the forward kinematics. This way our network would have a better knowledge of the hand model. Basically the general idea is to already impose a structure on our network like hand connectedness, thickness, elongation of our hand. We would be using point cloud for our predicted shape.

Milestone progress plan:

- Preprocess the dataset and finish experimenting, benchmarking and implementing different VAEs. 17.05.2026 or 24.05.2026.
- Find the right architecture for the diffusion model and train it on a small dataset. 14.06.2026
- Train our model on a larger dataset. Benchmark our model in speed and accuracy in comparison with similar models. Measure the robustness, try input dropout, input robustness. Benchmarking with different VAEs. 05.07.2026

References :

- [1] Jiaxin Lu, Hao Kang, Haoxiang Li, Bo Liu, Yiding Yang, Qixing Huang, Gang Hua. UGG: Unified Generative Grasping. arXiv:2311.16917
- [2] In Cho, Youngbeom Yoo, Subin Jeon, Seon Joo Kim. COD-VAE Representing 3D Shapes With 64 Latent Vectors for 3D Diffusion Models. arXiv:2503.08737
- [3] Xiaohui Zeng, Arash Vahdat, Francis Williams, Zan Gojcic, Or Litany, Sanja Fidler, Karsten Kreis. LION Latent Point Diffusion Models for 3D Shape Generation. arXiv:2210.06978
- [4] Charles R. Qi, Li Yi, Hao Su, Leonidas J. Guibas. PointNet++ Deep Hierarchical Feature Learning on Point Sets in a Metric Space. arXiv:1706.02413
- [5] Junyu Chen, Han Cai, Junsong Chen, Enze Xie, Shang Yang, Haotian Tang, Muyang Li, Yao Lu, Song Han. Deep Compression Autoencoder for Efficient High-Resolution Diffusion Models. arXiv:2410.10733
- [6] Yaoqing Yang, Chen Feng, Yiru Shen, Dong Tian. FoldingNet Point Cloud Auto-encoder via Deep Grid Deformation. arXiv:1712.07262
- [7] Gabriele Mario Caddeo, Pasquale Marra, Lorenzo Natale. Physically Grounded 3D Generative Reconstruction under Hand Occlusion using Proprioception and Multi-Contact Touch. arXiv:2604.09100
- [8] Yuanbo Wang, Zhaoxuan Zhang, Jiajin Qiu, Dilong Sun, Zhengyu Meng, Xiaopeng Wei, Xin Yang. Touch2Shape: Touch-Conditioned 3D Diffusion for Shape Exploration and Reconstruction. arXiv:2505.13091
- [9] Langzhe Gu, Hung-Jui Huang, Mohamad Qadri, Michael Kaess, Wenzhen Yuan. TouchAnything: Diffusion-Guided 3D Reconstruction from Sparse Robot Touches. arXiv:2604.08945